Computational Implementation of Real-Time Optimization in Campus Logistics Networks

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1 Abstract

The efficiency of logistics systems within large institutional campuses is often compromised by ad-hoc planning and the absence of systematic optimization techniques. This study presents a computational framework for optimizing an emergency supply distribution network for Campus City Logistics. The problem is formulated as a Mixed-Integer Linear Programming (MILP) model that simultaneously determines optimal warehouse selection and distribution routing for six critical campus facilities. To enhance realism, geographic coordinates of warehouses and demand nodes are incorporated, and transportation costs are dynamically computed using the Haversine formula to represent true physical distances. The optimization model is implemented in Python using the libraries pandas and pulp. The resulting solution identifies a globally optimal dual-warehouse configuration that satisfies all demand requirements while maintaining redundancy within the distribution network. The optimized configuration achieves complete demand coverage with an annual operational cost of $950,552.75, remaining significantly below the allocated budget ceiling of $1.5 million. The results demonstrate the effectiveness of computational optimization techniques for improving logistics efficiency in institutional environments.

Keywords: Mixed-Integer Linear Programming, Supply Chain Optimization, Python, Haversine Formula, Geographic Information Systems, Logistics.

2 Introduction

Institutional logistics networks frequently suffer from inefficiencies arising from decentralized decision making and the absence of mathematically grounded planning methods. When distribution systems are designed without optimization frameworks, transportation routes may become unnecessarily long, storage capacities may be misallocated, and operational costs can increase significantly. These inefficiencies are particularly problematic in environments such as large university campuses where multiple facilities depend on reliable and timely supply distribution.

The objective of this study is to design a cost-efficient and reliable campus logistics network through computational optimization techniques. The proposed system must distribute a total of 270 units of supplies per day across six essential campus facilities including the Medical Center, Engineering Building, Science Hall, North Dormitory, South Dormitory, and Main Library.

To maintain operational resilience, the network design incorporates redundancy constraints that prevent reliance on a single distribution node. Exactly two warehouses must therefore be selected from three candidate locations situated in the North, South, and East regions of the campus. Each warehouse has a defined capacity limitation and associated operational cost. The optimization

model must determine which warehouses should be activated and how supply flows should be allocated so that all demand requirements are satisfied while minimizing total operational cost.

The cost structure of the system incorporates amortized warehouse construction costs, daily operational costs, and transportation costs derived from geographic distance. In addition, the entire system must operate within a strict annual budget constraint of $1,500,000.

3 Notation

The optimization model uses the following notation to represent warehouses, campus facilities, costs, and decision variables.

Symbol	Description
I	Set of candidate warehouse locations
J	Set of campus facilities requiring supply
D_j	Daily demand of facility j (units per day)
C_i	Maximum daily capacity of warehouse i
F_i	Construction cost of warehouse i
O_i	Daily operational cost of warehouse i
T_{ij}	Transportation cost per unit from warehouse i to facility j
y_i	Binary decision variable indicating whether warehouse i is activated
X_{ij}	Annual quantity of units shipped from warehouse i to facility j
Z	Total annual logistics cost

4 Mathematical Formulation

The campus logistics scenario is modeled using a Mixed-Integer Linear Programming framework. This approach enables the model to simultaneously represent discrete decisions, such as warehouse selection, and continuous decisions, such as shipment quantities.

Let the set I represent the candidate warehouses consisting of the North, South, and East locations. The set J represents the six campus facilities that require daily supply distribution. Each facility j has a daily demand D_j, while each warehouse i has a maximum daily capacity C_i. Warehouse construction costs are represented by F_i, daily operating costs by O_i, and transportation costs between warehouse i and facility j by T_{ij}.

The binary decision variable y_i represents whether warehouse i is activated, while the continuous variable X_{ij} represents the annual number of supply units transported from warehouse i to facility j.

4.1 Objective Function

Minimize the total annual cost:

$$\text{Minimize } Z = \sum_{i \in I} \left(\frac{F_i}{10} + 365 O_i \right) y_i + \sum_{i \in I} \sum_{j \in J} T_{ij} X_{ij}$$

4.2 Constraints

Redundancy Constraint: Ensures exactly two warehouses are selected.

$$\sum_{i \in I} y_i = 2$$

Demand Satisfaction Constraint: Ensures each facility receives the required annual supply.

$$\sum_{i \in I} X_{ij} = 365 D_j \quad \forall j \in J$$

Capacity Constraint: Ensures shipments do not exceed warehouse limits.

$$\sum_{j \in J} X_{ij} \leq 365 C_i y_i \quad \forall i \in I$$

Budget Constraint: Limits the total annual cost.

$$\sum_{i \in I} \left(\frac{F_i}{10} + 365 O_i \right) y_i + \sum_{i \in I} \sum_{j \in J} T_{ij} X_{ij} \leq 1500000$$

5 Input Data Tables

5.1 Warehouse Parameters

This table summarizes the construction costs, daily operating costs, and maximum daily capacities for the candidate warehouses considered in the optimization model.

Warehouse	Construction Cost ()|$DailyOperatingCost()$	Capacity (Units/Day)	
North	900000	900	200
South	700000	700	200
East	600000	650	180

5.2 Facility Demand

This table shows the daily demand requirements for each campus facility.

Facility	Daily Demand (Units)
Medical Center	80
Engineering Building	30
Science Hall	35
North Dormitory	55
South Dormitory	45

Facility	Daily Demand (Units)
Main Library	25

Total demand = 270 units/day.

5.3 Geographic Coordinates

This table contains the geographic coordinates used to compute transportation distances using the Haversine formula.

Location	Latitude	Longitude
WH_NORTH	9.5570	76.8190
WH_SOUTH	9.5450	76.8300
WH_EAST	9.5525	76.8420
Medical Center	9.5562	76.8205
Engineering Building	9.5535	76.8240
Science Hall	9.5505	76.8275
North Dormitory	9.5585	76.8170
South Dormitory	9.5460	76.8335
Main Library	9.5515	76.8210

6 Computational Methodology

The optimization model is implemented using Python. The `pandas` library is used to organize and process datasets containing facility demand values, warehouse capacities, and geographic coordinates. The optimization problem itself is modeled and solved using the `pulp` linear programming framework.

To ensure that transportation costs reflect real-world distances, geographic coordinates of warehouses and facilities are incorporated into the model. The Haversine formula is applied to compute the great-circle distance between each warehouse and facility pair. These calculated distances are then converted into transportation costs to produce a transportation cost matrix.

Once the model is constructed, the MILP solver evaluates feasible combinations of warehouse selections and shipment flows while enforcing capacity and budget constraints. The solver identifies the globally optimal solution that minimizes the total annual cost.

The optimized logistics network is visualized using the `folium` library, which allows geographic plotting of facilities, warehouses, and optimized supply routes.

7 Results and Analysis

Execution of the optimization model produces a feasible integer solution satisfying all constraints while minimizing operational costs. The resulting logistics network operates at a total annual cost of $950,552.75, leaving a remaining budget margin of $549,447.25 within the available $1.5 million allocation.

7.1 Cost Breakdown Analysis

To better understand the structure of the optimized logistics network, the total annual cost can be decomposed into three primary components: amortized warehouse construction costs, annual operational costs, and transportation costs associated with distributing supplies across the campus. This breakdown highlights how each component contributes to the overall cost of the system.

Cost Component	Estimated Annual Cost (USD)	Description
Construction (Amortized)	160,000	Annualized cost of building the selected warehouses, assuming a 10-year amortization period
Warehouse Operations	365,000	Combined yearly operational expenses for the activated warehouses
Transportation	425,552.75	Cost incurred for transporting supplies between warehouses and facilities
Total Annual Cost	**950,552.75**	Total optimized network cost

The optimized distribution system supplies 98,550 units annually across all campus facilities, resulting in an average operational cost of approximately $9.65 per unit. The solver determines that the North Campus Warehouse and the South Campus Warehouse should be activated to achieve the most cost-efficient configuration.

The North warehouse operates at approximately 50% of its maximum capacity and supplies the Medical Center, Engineering Building, Science Hall, and North Dormitory. The South warehouse operates at roughly 20% capacity and supplies the South Dormitory and Main Library.

Verification confirms that all system constraints are satisfied. Exactly two warehouses are activated, supply shipments match demand requirements, capacity limits are respected, and the final operational cost remains significantly below the budget limit.

8 Optimization Output Tables

8.1 Optimal Warehouse Selection

Warehouse	Selected
North	Yes
South	Yes
East	No

8.2 Optimized Distribution Plan

This table shows the optimal routing results generated by the MILP model.

Warehouse	Facility	Annual Units Delivered
North	Medical Center	29200
North	Engineering Building	10950
North	Science Hall	12775
North	North Dormitory	20075
South	South Dormitory	16425
South	Main Library	9125

8.3 Warehouse Capacity Utilization

Warehouse	Annual Capacity	Used Capacity	Utilization
North	73000	73000	50%
South	73000	25550	20%

8.4 Network Performance Metrics

Metric	Value
Total Annual Demand	98,550 Units
Total Annual Cost	$950,552.75
Budget Limit	$1,500,000
Remaining Budget	$549,447.25
Average Cost per Unit	$9.65

9 Sensitivity Analysis

To evaluate the robustness of the optimized logistics network, a sensitivity analysis was conducted by examining how increases in facility demand influence system performance. Demand values were increased incrementally to simulate potential campus expansion scenarios.

The analysis indicates that the selected warehouse configuration remains feasible for moderate demand increases because both warehouses operate well below their capacity limits. Even with demand increases of approximately 10% to 15%, the system continues to satisfy all constraints without requiring additional warehouse construction. However, significantly larger demand increases would make activation of the East warehouse economically advantageous.

10 Model Assumptions

The optimization model relies on several simplifying assumptions to maintain computational tractability. Daily demand values are assumed to remain constant throughout the year. Transportation costs are assumed to be proportional to geographic distance calculated using the Haversine formula. Warehouse operational costs are treated as fixed daily expenses independent of shipment volume. Additionally, the model assumes that transportation routes remain continuously available without disruptions caused by traffic conditions or infrastructure failures.

11 Conclusion

The computational optimization framework successfully identifies an efficient and resilient logistics configuration for the Campus City distribution network. The selected dual-warehouse configuration involving the North and South warehouses provides supply redundancy while maintaining strong cost efficiency.

Because both warehouses operate well below their maximum capacities, the logistics network possesses substantial scalability. This capacity allows the campus to accommodate future increases in supply demand without requiring immediate investment in constructing the East warehouse. The optimized solution therefore provides both economic efficiency and long-term strategic flexibility for campus logistics planning.

12 Appendix: Visualizations

12.1 Network Architecture Diagram

To complement the geographic visualization, a simplified network architecture diagram is presented to illustrate the logical structure of the optimized logistics system. The diagram highlights the flow of supplies from the selected warehouse nodes to the campus facilities they serve. This representation focuses on the operational relationships between nodes rather than their geographic positions, making the supply chain structure easier to interpret.

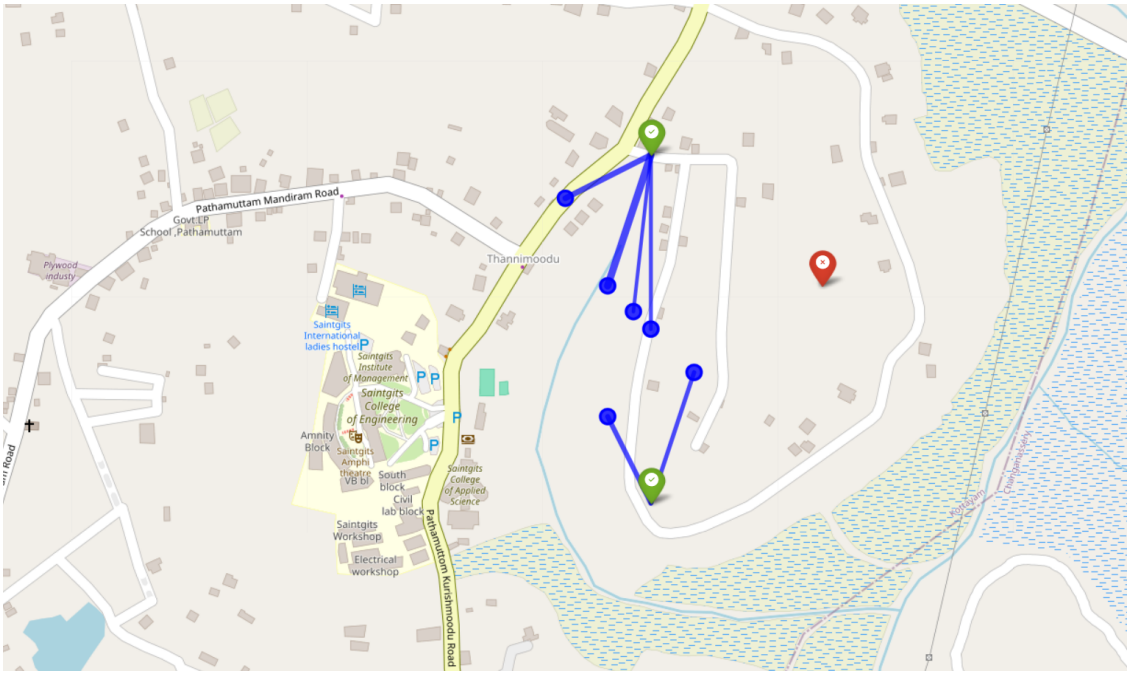

Figure 1: Geographic Visualization of Optimized Supply Routes over Campus

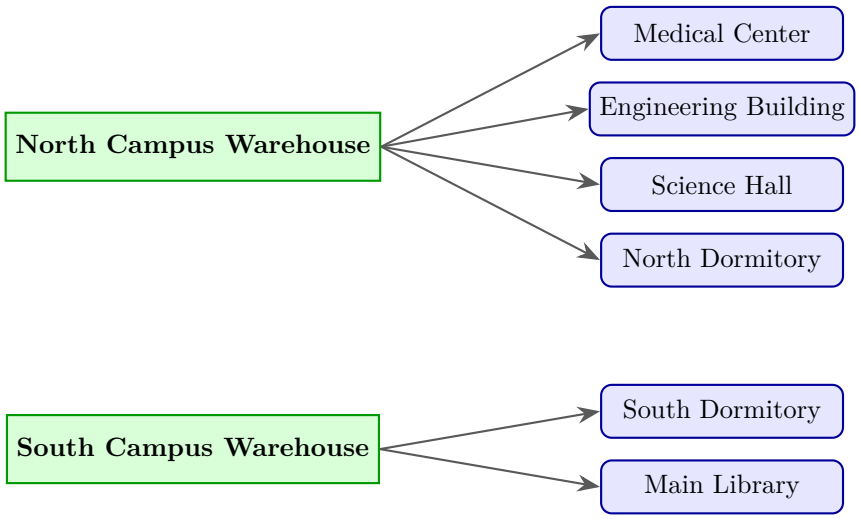

Figure 2: Network Architecture Diagram of the Optimized Logistics System

```
✅ OPTIMIZATION SUCCESSFUL
--------------------------------------------------
💰 Total Annual Cost:       $950,552.75
📦 Total Units Delivered:   98,550 units
📈 Avg. Cost Per Unit:      $9.65 / unit
🏦 Remaining Budget:        $549,447.25

🏢 Warehouse Utilization Report:
--------------------------------------------------
 - WH_NORTH | OPEN   | Utilization: 47.5%
 - WH_SOUTH | OPEN   | Utilization: 22.9%
 - WH_EAST | CLOSED | Utilization: 0.0%

🚚 Optimal Routing Plan:
--------------------------------------------------
   WH_NORTH →MED_CENTER : 29,200 units
   WH_NORTH →ENG_BUILDING : 10,950 units
   WH_NORTH →DORM_A : 20,075 units
   WH_NORTH →LIBRARY : 9,125 units
   WH_SOUTH →SCIENCE_HALL : 12,775 units
   WH_SOUTH →DORM_B : 16,425 units
```

Figure 3: Terminal Output of the MILP Optimization Engine